AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 13 · STUDENT LESSON

Multimodal Evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 12: different modalities can contain different evidence.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 13: Multimodal **Evidence**

Multiple modalities help only when their information is aligned and relevant.



CREATOR MISSION

Design a plant helper using image plus question text.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Image generation and provenance. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can combine **evidence** from modalities without treating hidden content as visible.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Multimodal Evidence case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use multimodal **evidence** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

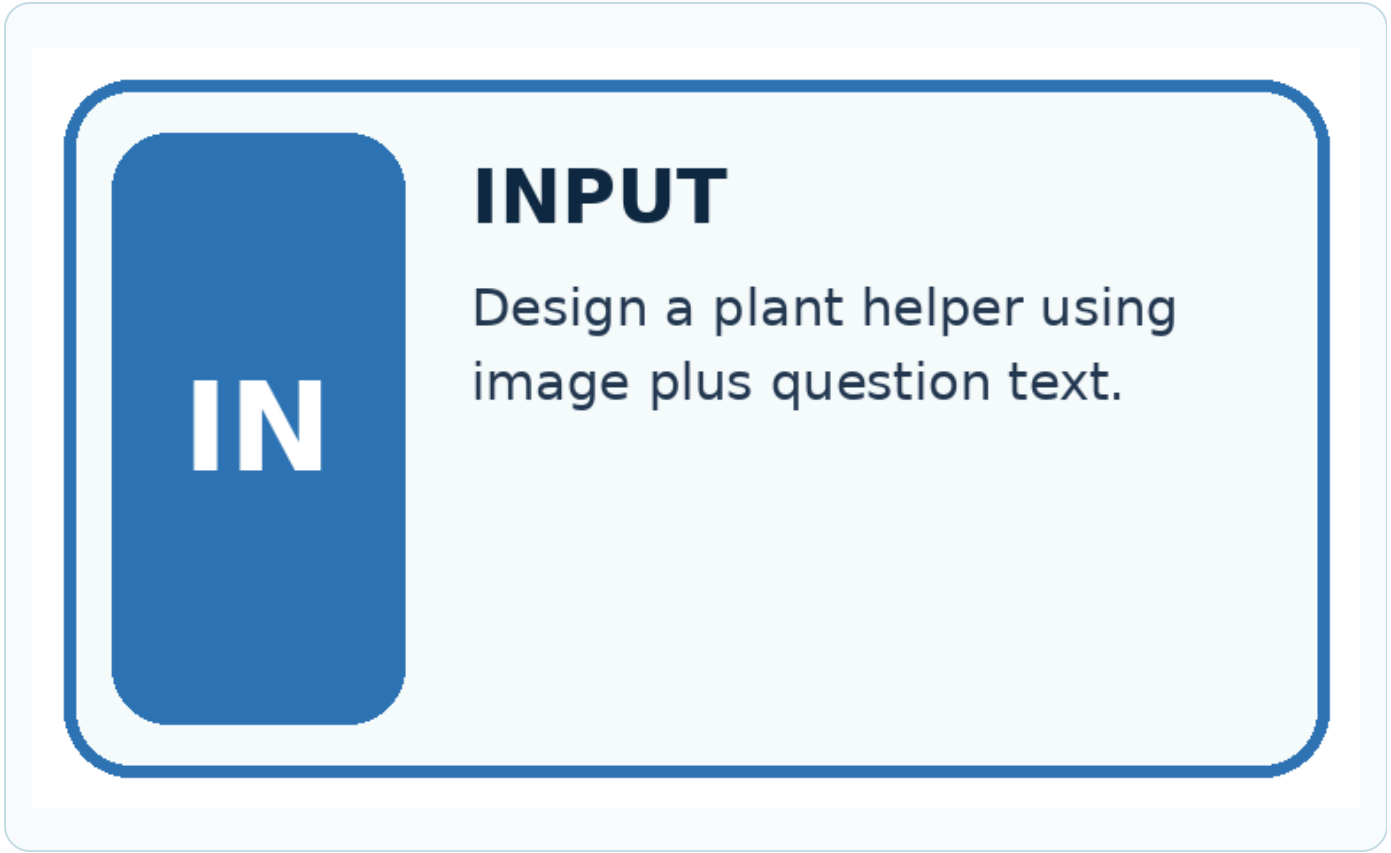
See how it works

> Follow the mechanism for Multimodal Evidence. Connect each labelled input, deciding step and result.

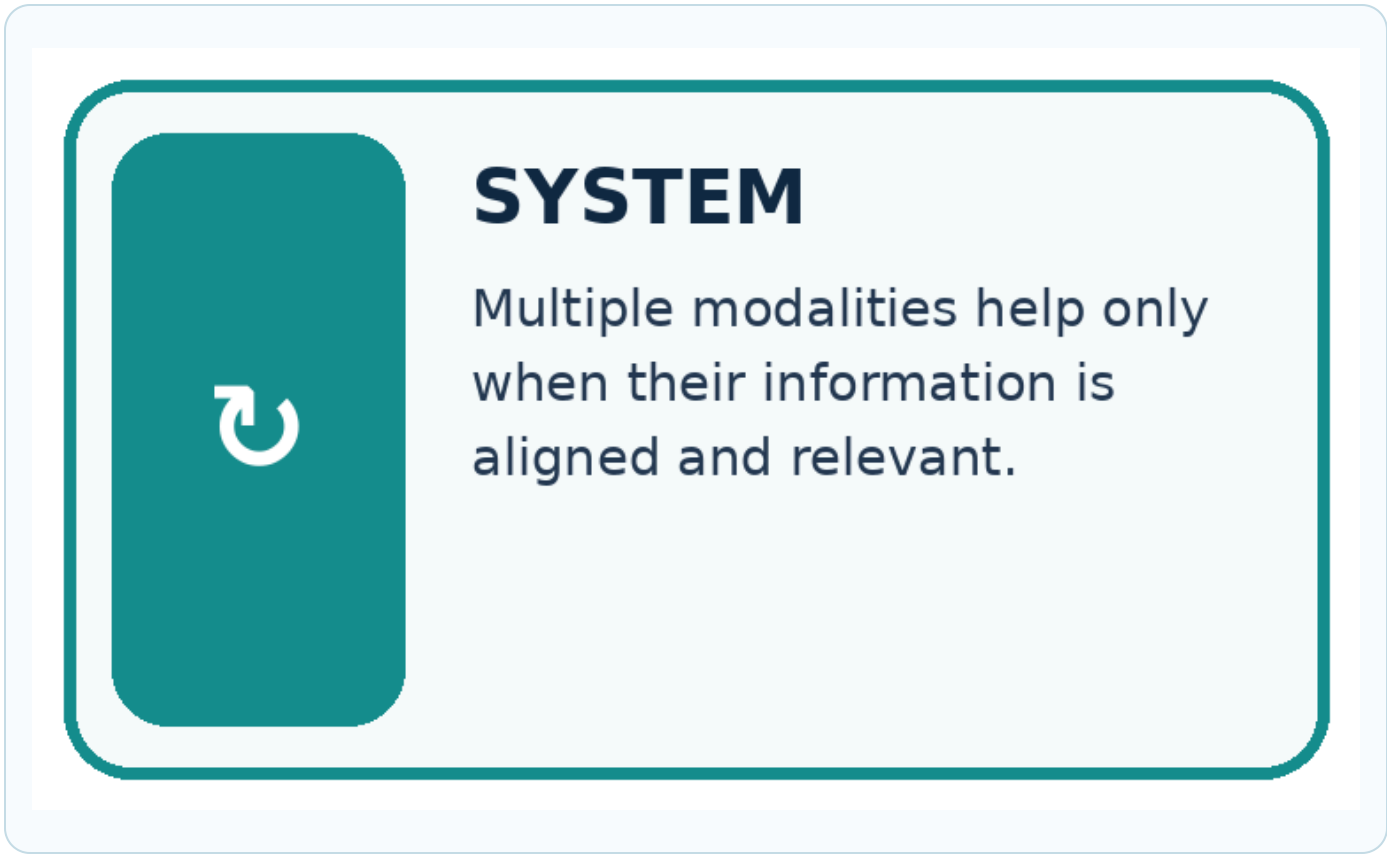
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Multiple modalities help only when their information is aligned and relevant.

↓


OUT
EVIDENCE
Design a plant helper using image plus question text.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare **conflict**ing **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Speech pipelines and multimodal evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.	Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.	There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

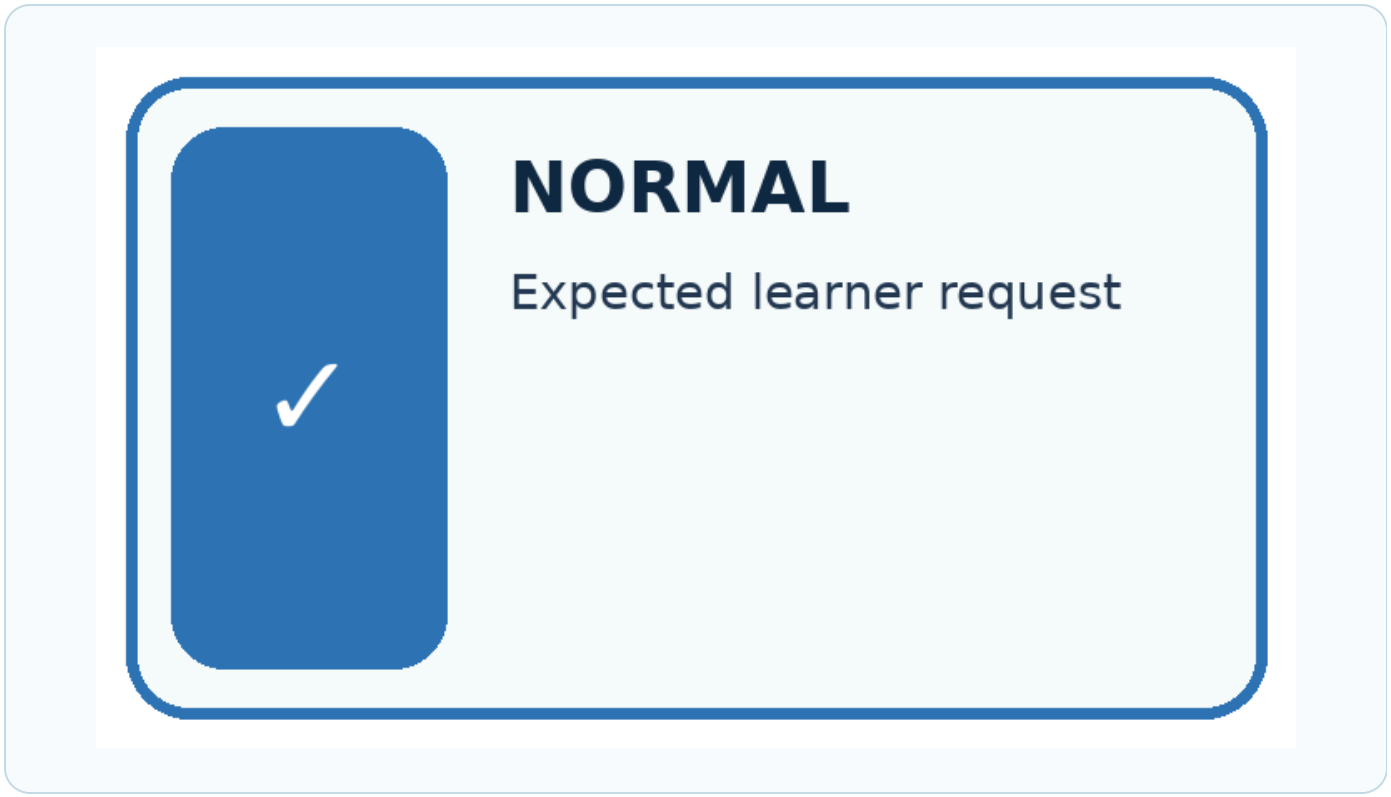
> Compare the supplied cases for Multimodal Evidence. Identify the detail that changes the conclusion.

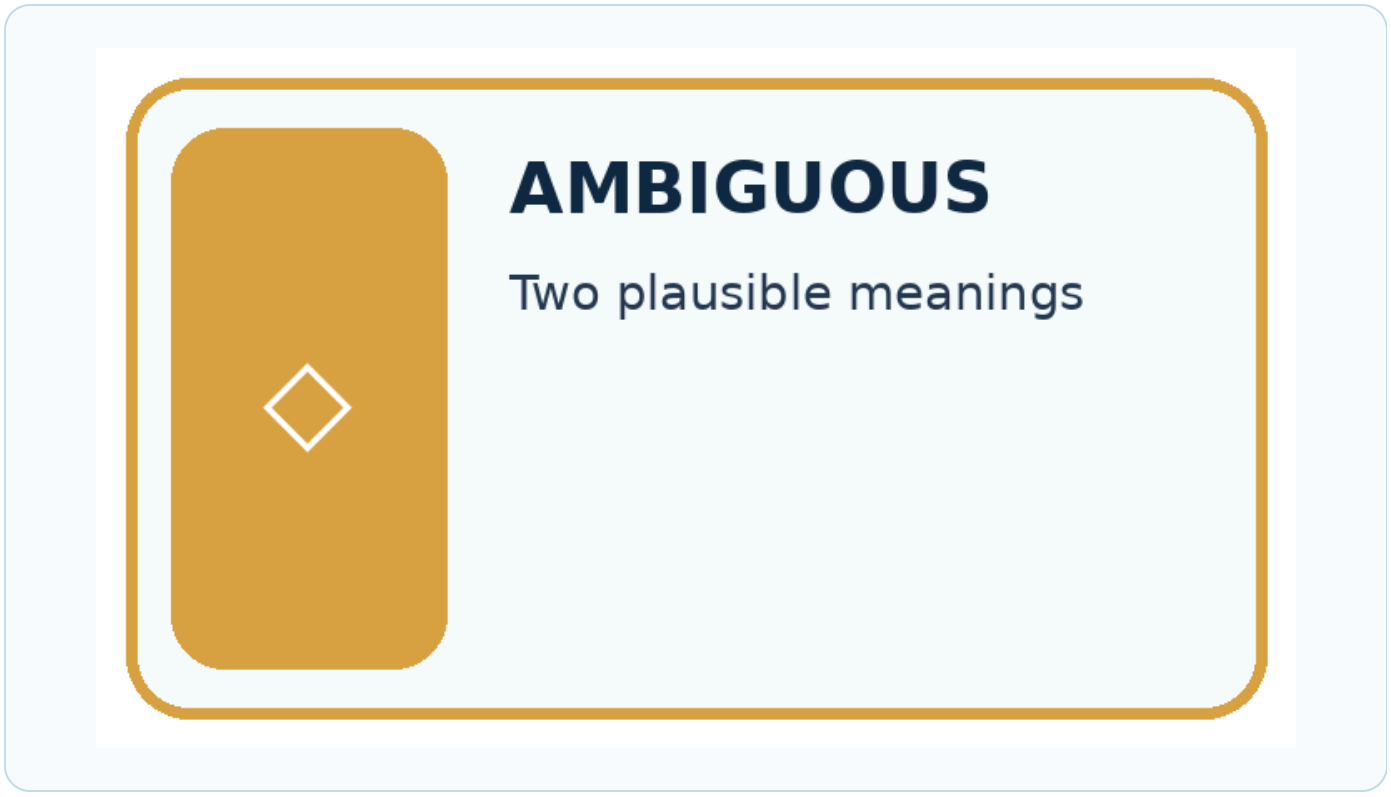
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations.

Predicted result

Prediction: The total is now supported as five.

Why the result follows

Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

STUDENT LESSON · 5 OF 18

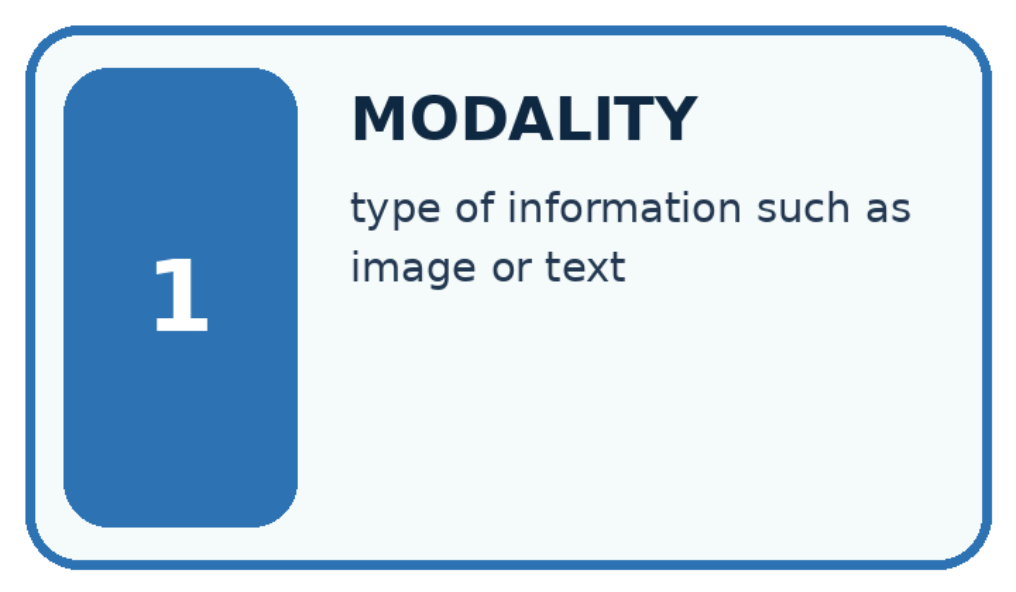
Words that unlock the idea

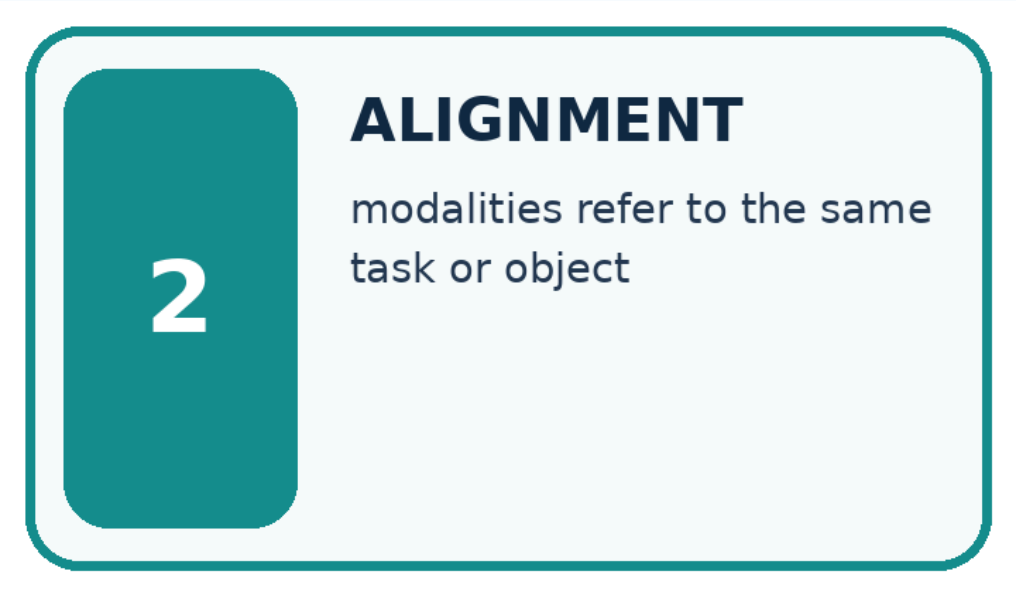
> Use the vocabulary of Multimodal Evidence to explain a concrete decision, rather than only memorising definitions.

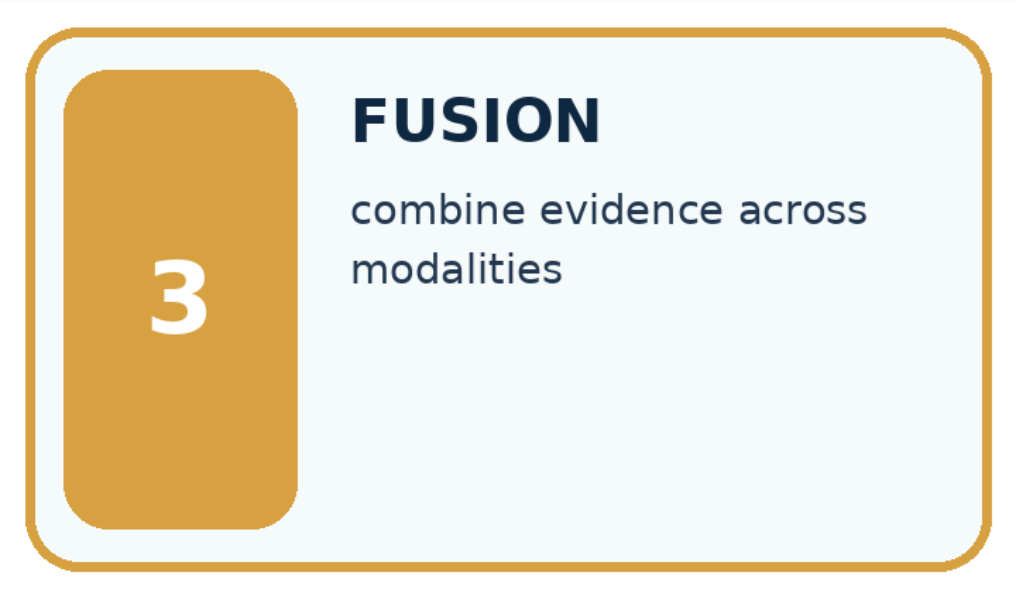
VOCABULARY LAB

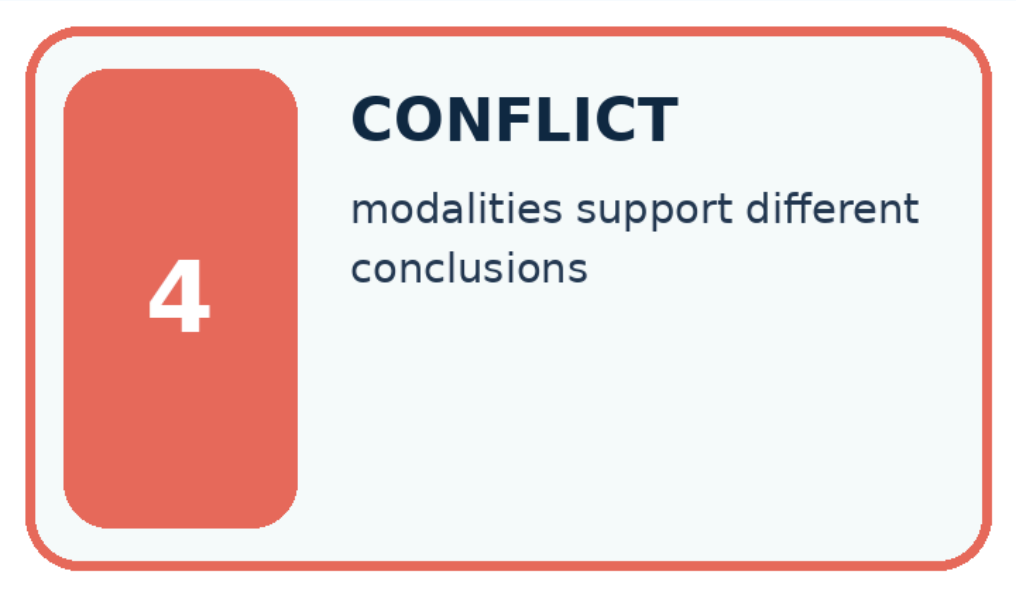
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
MODALITY
type of information such as image or text


2
ALIGNMENT
modalities refer to the same task or object





Word	My own example	Not this / boundary
modality	A form of information, such as text, image, audio or video.	
alignment	How well a system's behaviour matches the intended goals, constraints and values in a stated setting.	
fusion	Combining information from multiple sources or modalities.	

Word	My own example	Not this / boundary
conflict	A disagreement between instructions, sources, labels or requirements.	

Words used in the closer explanation

Word	Meaning
Transcription	A text representation of speech
Error propagation	An earlier error affecting later processing

---

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Multimodal Evidence, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 13 • WORKED EVIDENCE

1. Image shows leaf spots; text asks about watering history.
2. Each modality supplies different relevant evidence.
3. If photo and caption describe different plants, fusion is unsafe.
4. Check alignment, source, and uncertainty before combining.

1. Image shows leaf spots; text asks about watering history.

2. Each **modality** supplies different relevant **evidence**.

3. If photo and caption describe different plants, **fusion** is unsafe.

4. Check **alignment**, source, and uncertainty before combining.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity **conflict** and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Worked explanation · Multimodal Evidence

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Multimodal Evidence

A fictional Study Buddy audits a desk description and a separate caption.

The facts we will use

Supplied facts

Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Supplied facts

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

AI ACADEMY / AI-3/13

Multimodal Evidence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Visible evidence

Three pencils are explicitly visible in the open box.

2. Hidden region

The closed box's pencil count is unknown.

3. Caption claim

Five total requires two additional pencils if these are the only relevant locations.

4. Supported response

At least three are supported; total five is not established by the description.

CHANGE ONE THING

Open the closed box and a verified observation shows two pencils inside, with no other desk locations.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Visible evidence	Three pencils are explicitly visible in the open box.
2. Hidden region	The closed box’s pencil count is unknown.
3. Caption claim	Five total requires two additional pencils if these are the only relevant locations.
4. Supported response	At least three are supported; total five is not established by the description.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

What this example does not establish

Keep the stated limit

The conclusion depends on the supplied description’s accuracy; no actual image inspection or exhaustive room inventory was performed.

Keep the stated limit

Read the labelled picture: Why were both “exactly three” and “exactly five” unsupported before opening the box?

Keep the stated limit

Change one thing: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Multimodal Evidence claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Adding more modalities always improves accuracy. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

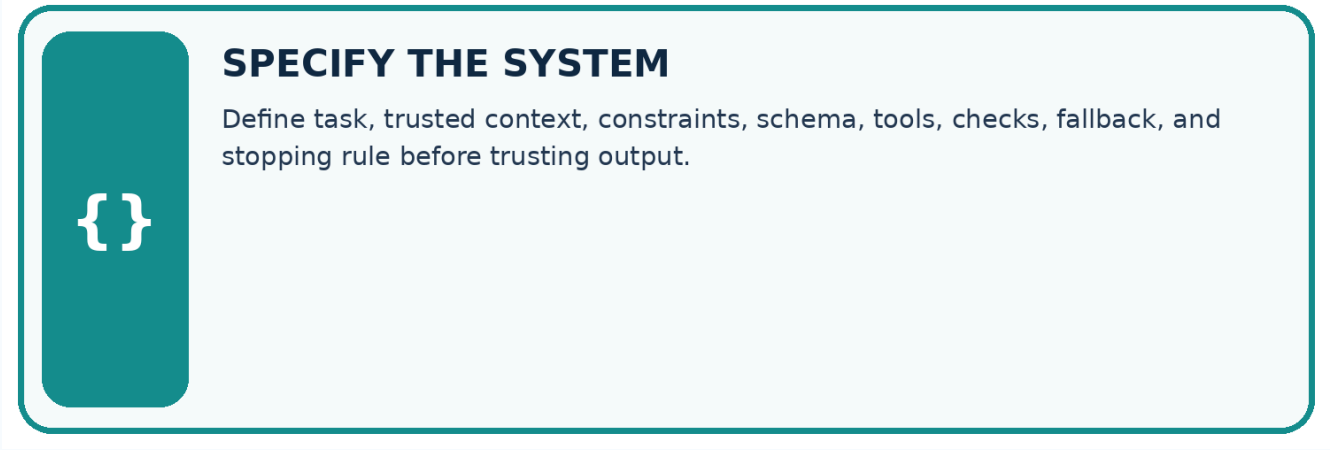
Practise together

> Practise Multimodal Evidence with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Speech pipelines and multimodal **evidence**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why were both “exactly three” and “exactly five” unsupported before opening the box?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Multimodal Evidence. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Match image/text pairs, flag irrelevant or **conflict**ing **evidence**, and design ask-more or abstain behavior.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

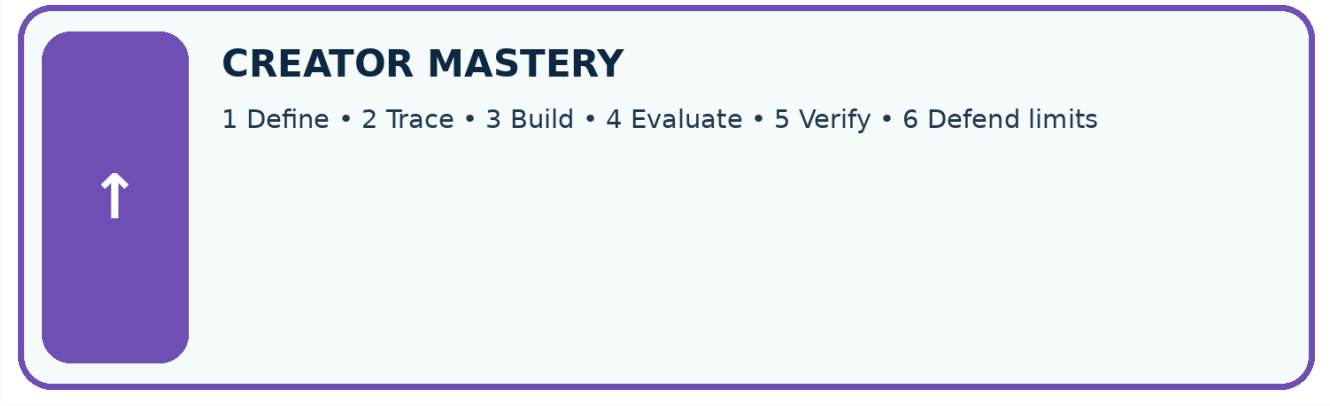
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Multimodal Evidence. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

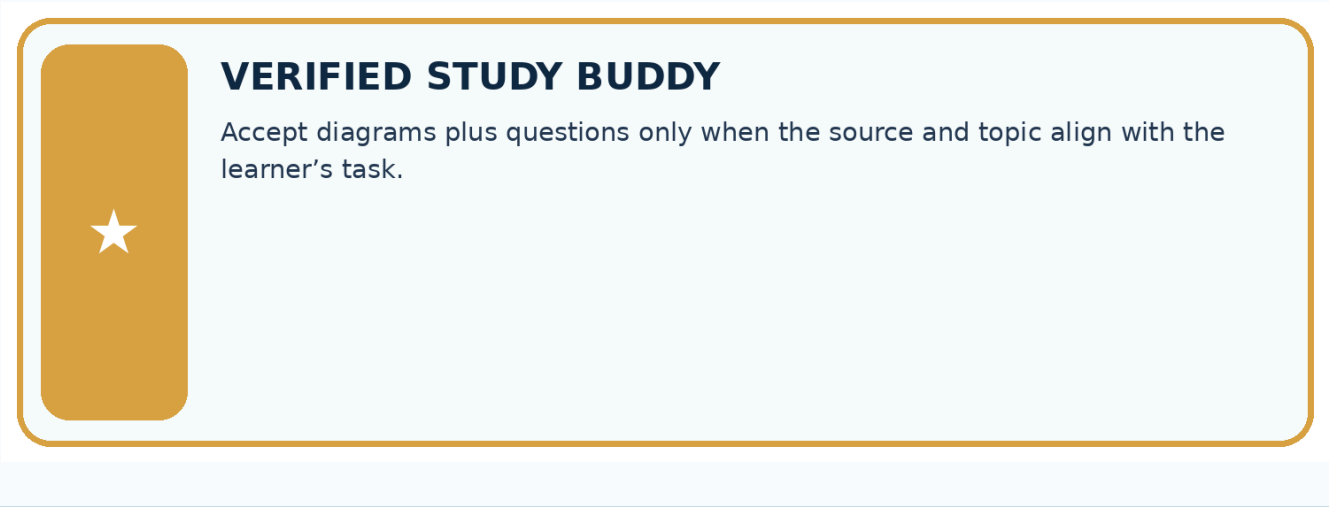
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Accept diagrams plus questions only when the source and topic align with the learner's task.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information.

STUDENT LESSON · 12 OF 18

Show what you know

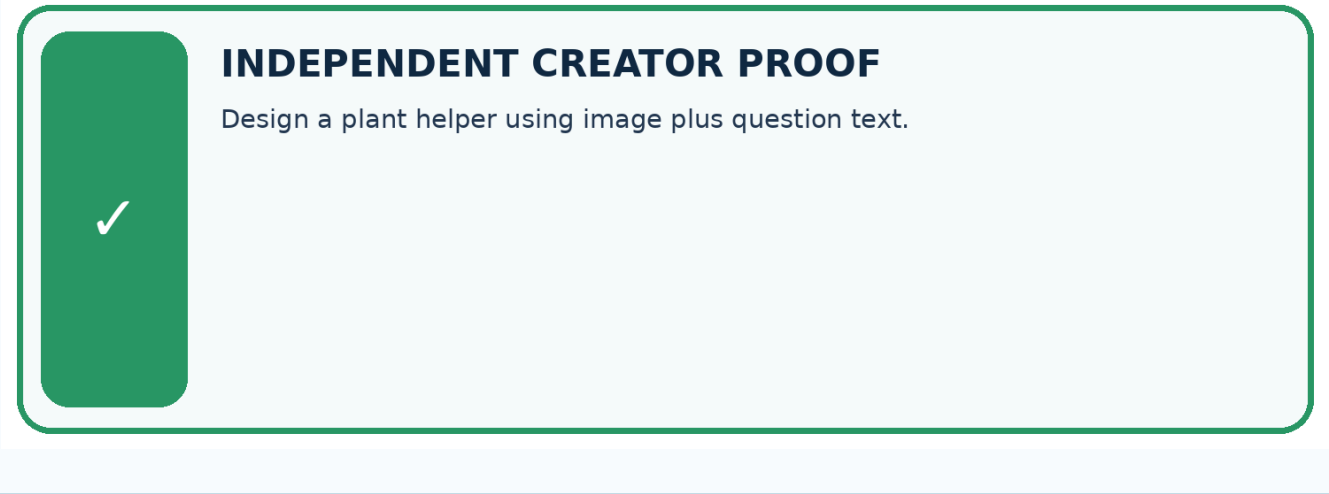
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Design a plant helper using image plus question text.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Multimodal Evidence case for supported practice. Repeated examples do not become fresh assessments.

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An image shows three plants; caption claims a fourth outside the frame.

The question

> What can the image establish?

Worked explanation

Three visible plants. It cannot settle an off-frame claim. Attribute claims and preserve missing **evidence**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Separate visible and off-frame claims

Evidence or stage	Value or result
Image	Three visible plants
Caption	A fourth outside the frame
Evidence limit	Image cannot settle the off-frame claim

An image shows three plants; caption claims a fourth outside the frame.

> What can the image establish?

Three visible plants. It cannot settle an off-frame claim. Attribute claims and preserve missing **evidence**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

> A description reports two visible plants and a third opaque pot whose contents cannot be seen. A caption says “three plants”. What is supported?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A chart image shows 12 plants; its caption says 21. What should a multimodal assistant report before drawing a conclusion?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Accept diagrams plus questions only when the source and topic align with the learner's task.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A caption says four seedlings; the image shows three and an opaque box. Is a fourth seedling disproved? Write an **evidence**-aware response.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Multimodal Evidence, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional Study Buddy audits a desk description and a separate caption.

By the end, I can	How I will show it
Explain the mechanism and terminology of speech pipelines and multimodal evidence	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Multimodal Evidence without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy audits a desk description and a separate caption.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At least three are supported; total five is not established by the description.

1. Caption claim

2. Supported response

3. Hidden region

4. Visible **evidence**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Supported response At least three are supported; total five is not established by the description.

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The total is now supported as five.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The total is now supported as five. 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Repair a missing building block

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and provenance.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

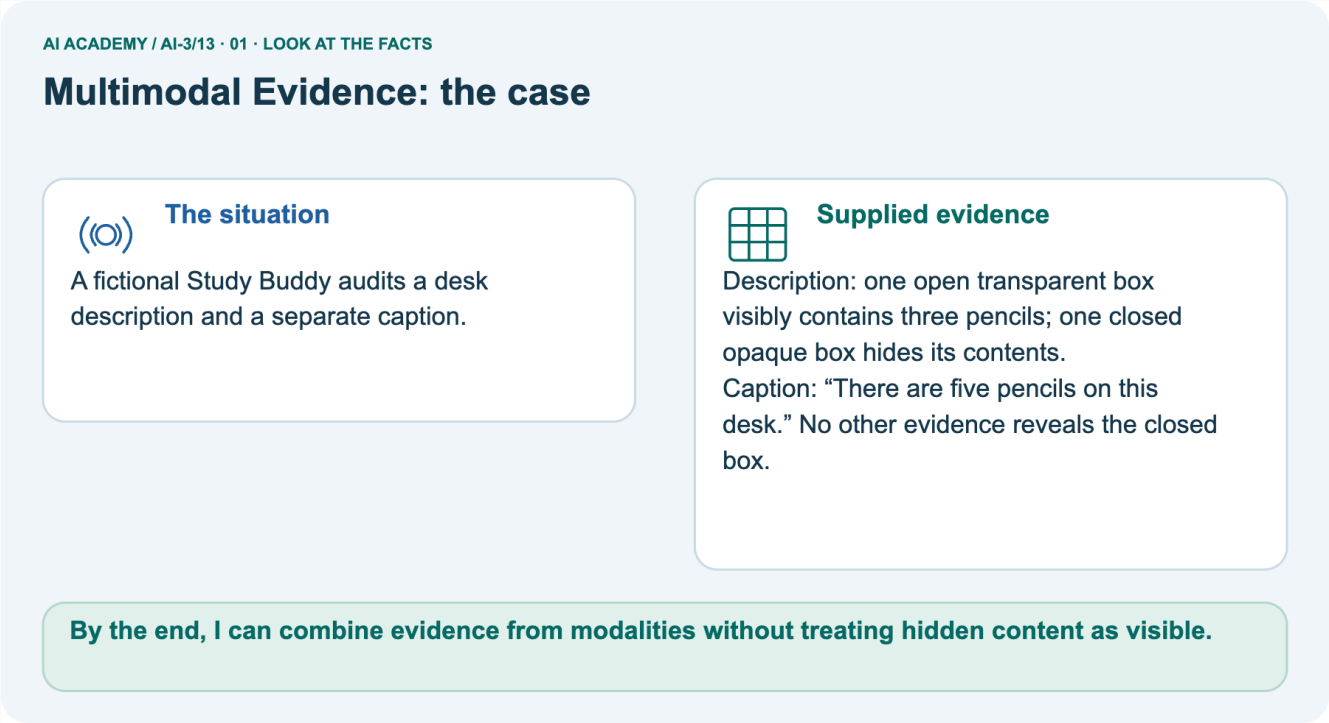
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Multimodal Evidence graphic using its labels, deciding evidence and relevant limitation.

01. Multimodal Evidence: the case



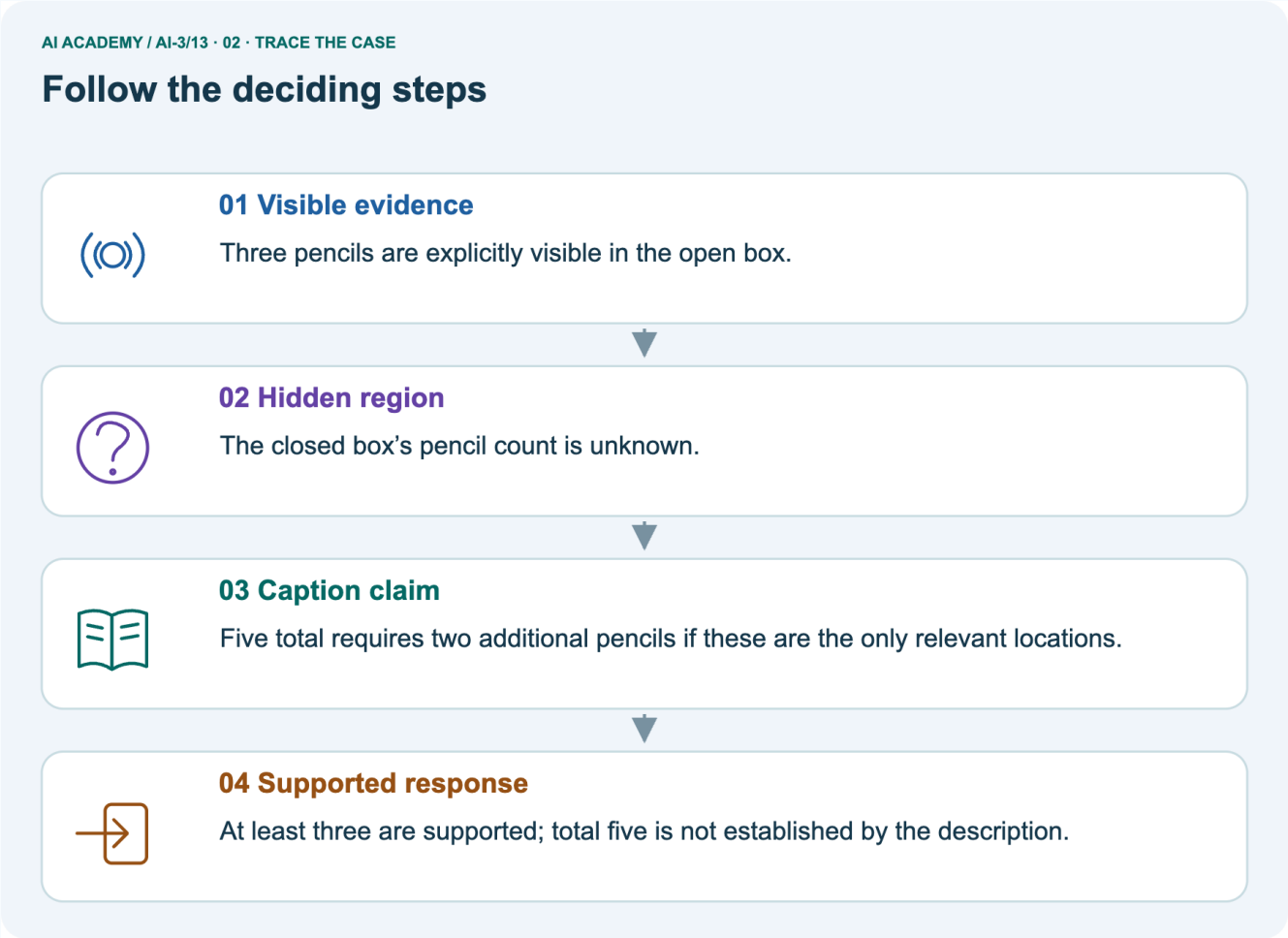
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy audits a desk description and a separate caption. Supplied facts: Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

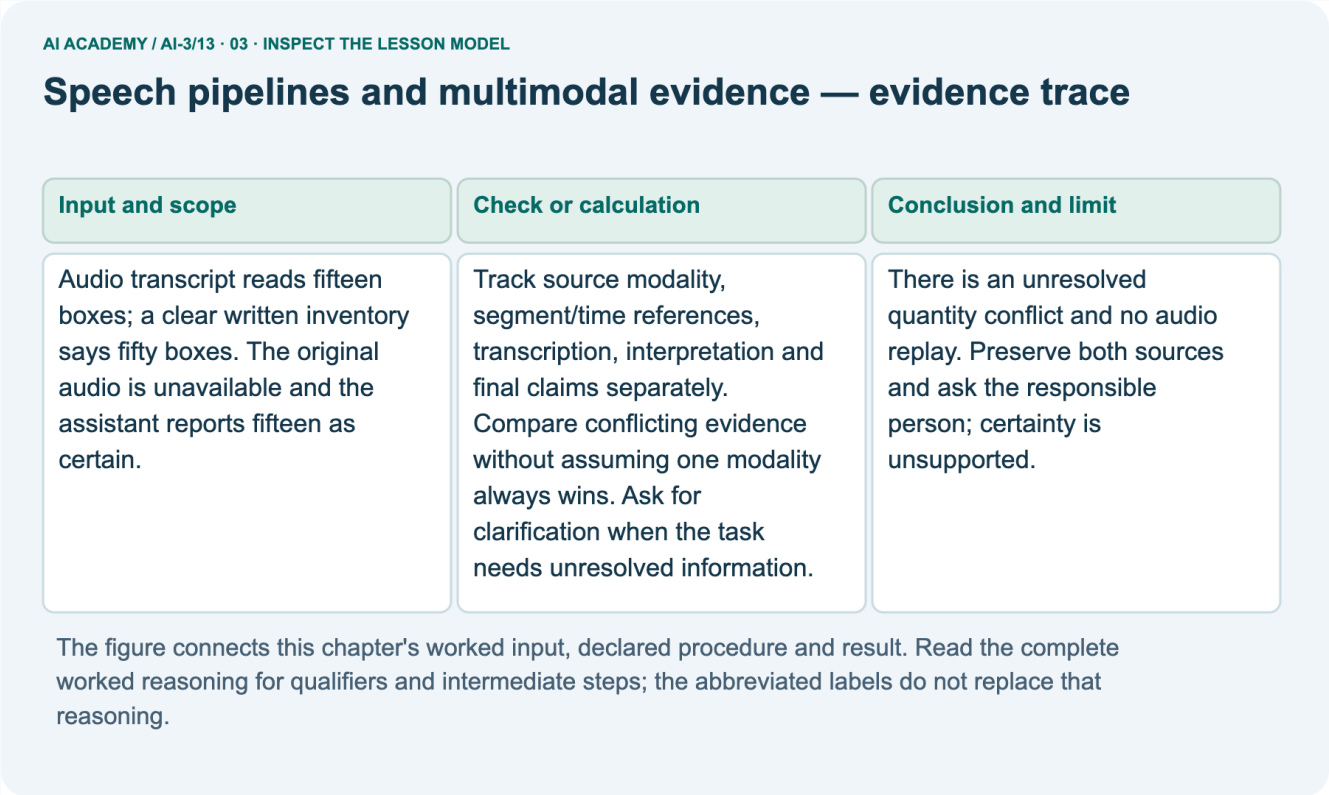


Follow the labelled picture with the matching explanation. Open the vector diagram

Visible **evidence**: Three pencils are explicitly visible in the open box. → Hidden region: The closed box's pencil count is unknown. → Caption claim: Five total requires two additional pencils if these are the only relevant locations. → Supported response: At least three are supported; total five is not established by the description.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

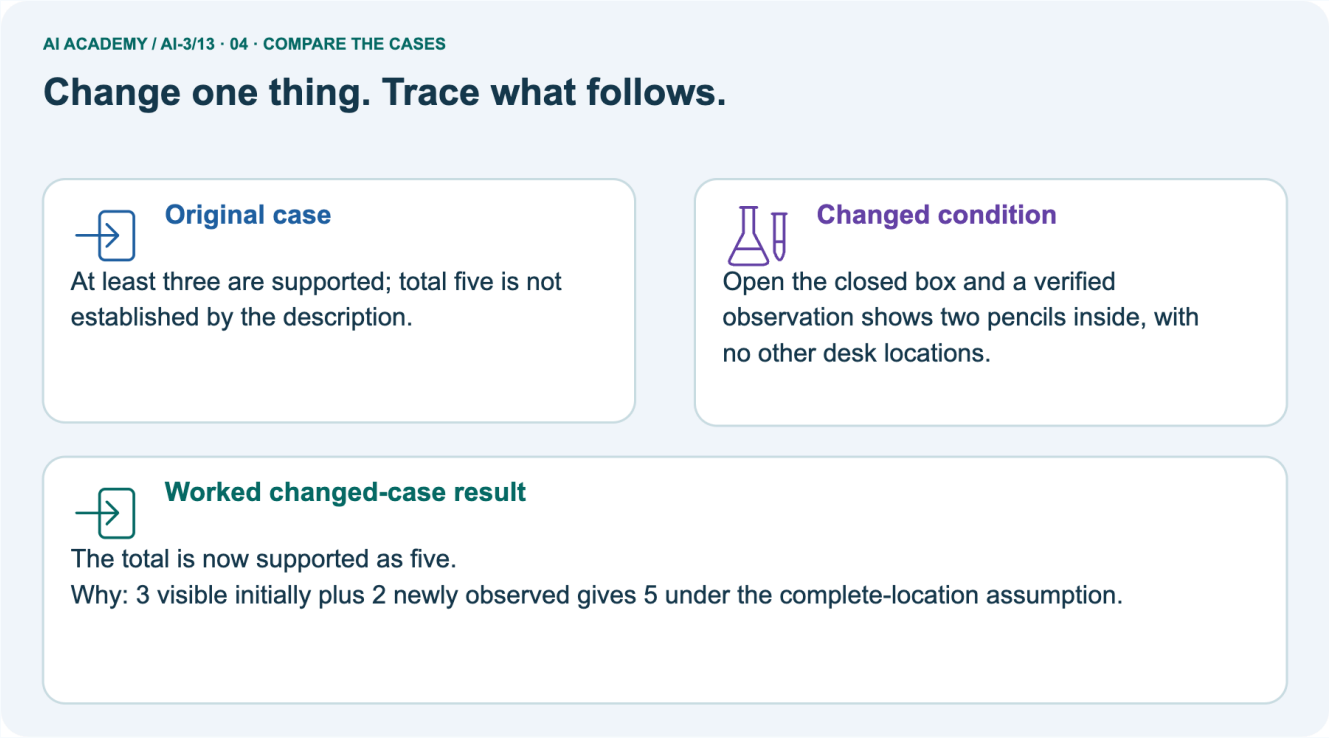
The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare **conflict**ing **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity **conflict** and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



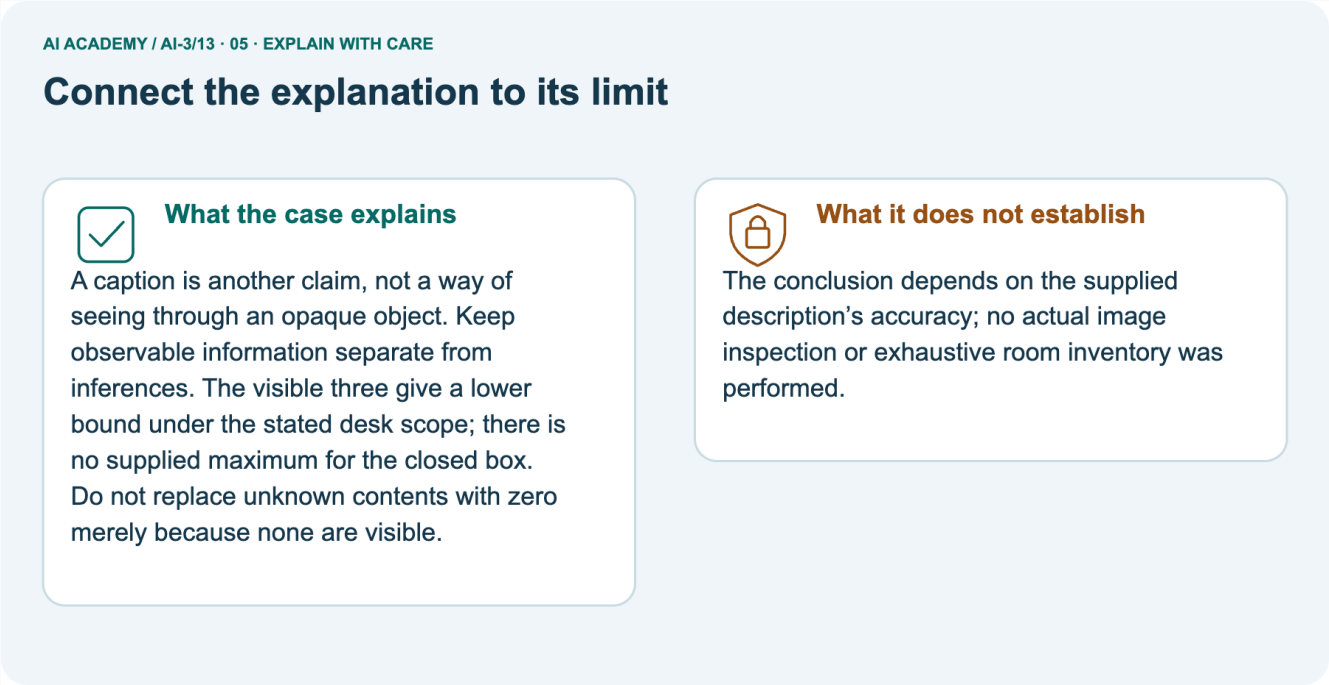
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At least three are supported; total five is not established by the description. Changed condition: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations.

Expected result: The total is now supported as five. Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

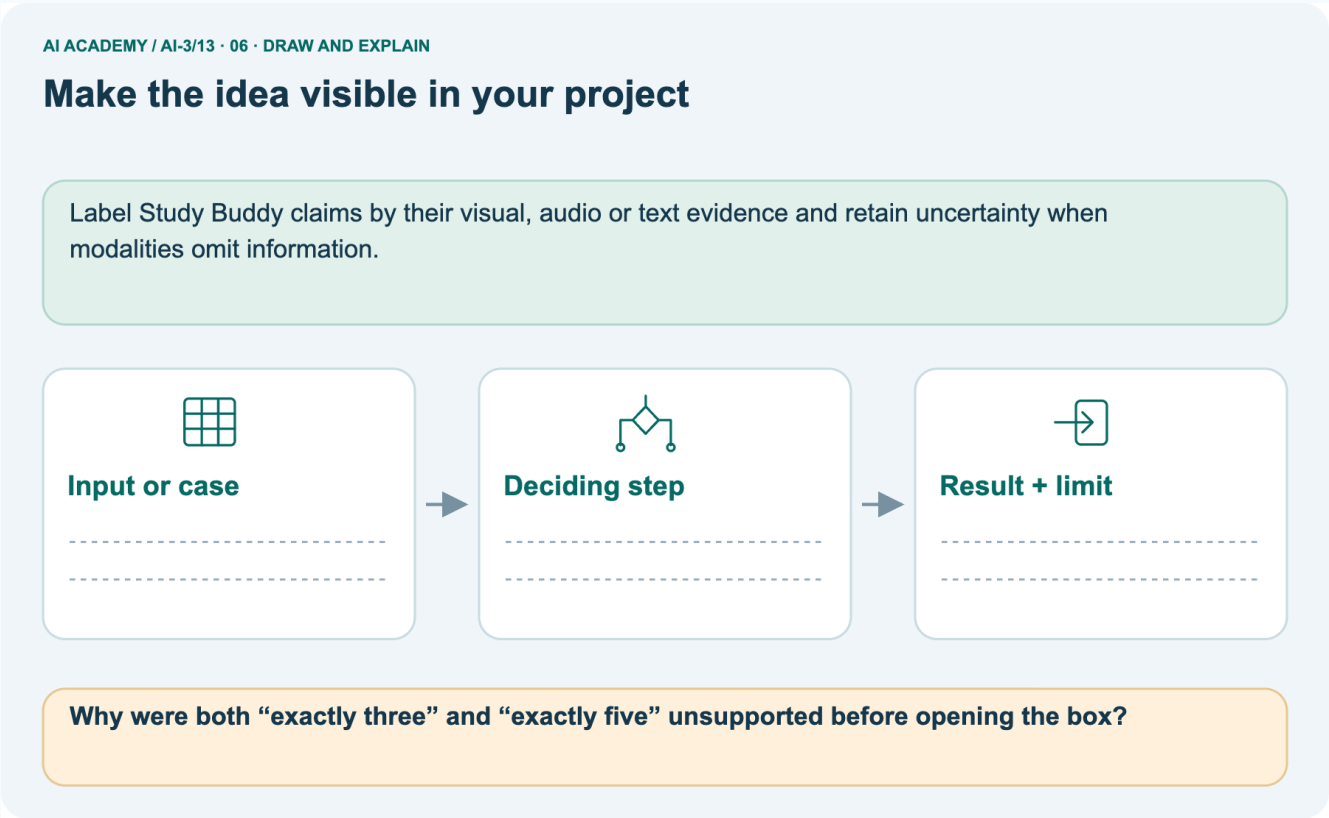
A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Limit: The conclusion depends on the supplied description’s accuracy; no actual image inspection or exhaustive room inventory was performed.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information. Draw your **input**, deciding step, result and limit.

Supported question: Why were both “exactly three” and “exactly five” unsupported before opening the box?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.